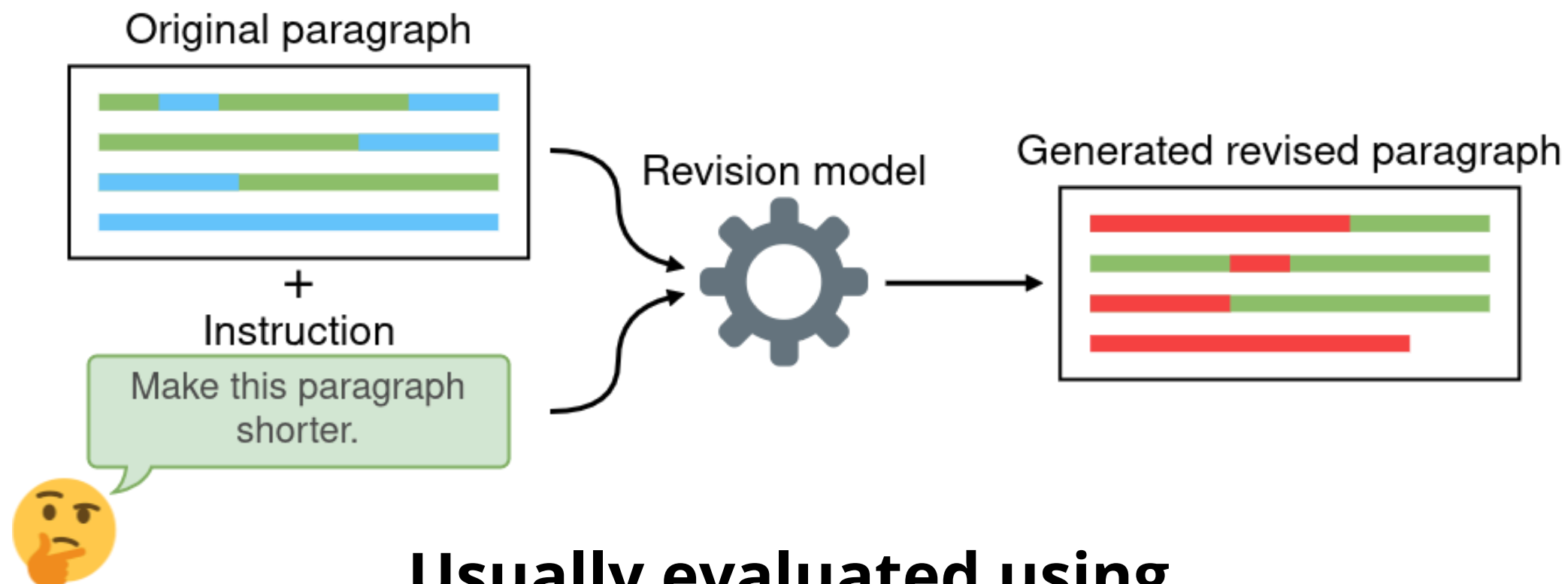


Identifying Reliable Evaluation Metrics for Scientific Text Revision

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Context - Task

Paragraph revision task



Usually evaluated using

Similarity based on:

- n-grams

- embeddings



BLEU
ROUGE
METEOR
SARI
GLEU
BERTScore



Input:

- Gold + Generated
- Original + Gold + Generated

Context - Dataset

ParaRev¹ test set

- 258 pairs of revised paragraphs
- 2 annotations/paragraph
 - = 516 datapoints
- 5 types of revision

Written by the authors of
the paper.
Collected on Open Review.

Previous annotation task
Not the original author

Original paragraph

[...] **Nevertheless, challenges exist for** developing deep learning-based models to predict mutational effects on protein-protein **binding. The major challenge is the scarcity of experimental data —** only a few **thousands of** protein **mutations** annotated with **the change** in binding **affinity** are publicly available (Geng et al., **2019b**). **This hinders** supervised learning **as the insufficiency of training data tends to cause over-fitting.** [...]

Gold revised paragraph

[...] **However,** developing deep learning-based models to predict mutational effects on protein-protein **binding is challenging due to the scarcity of experimental data. Only** a few **thousand** protein **mutations,** annotated with **changes** in binding **affinity,** are publicly available (Geng et al., **2019b**), **making** supervised learning **challenging due to the potential for overfitting with insufficient training data.** [...]

Manual Annotation

Revision Instruction

Rewrite this paragraph for better readability.

Intention label

Rewriting_medium

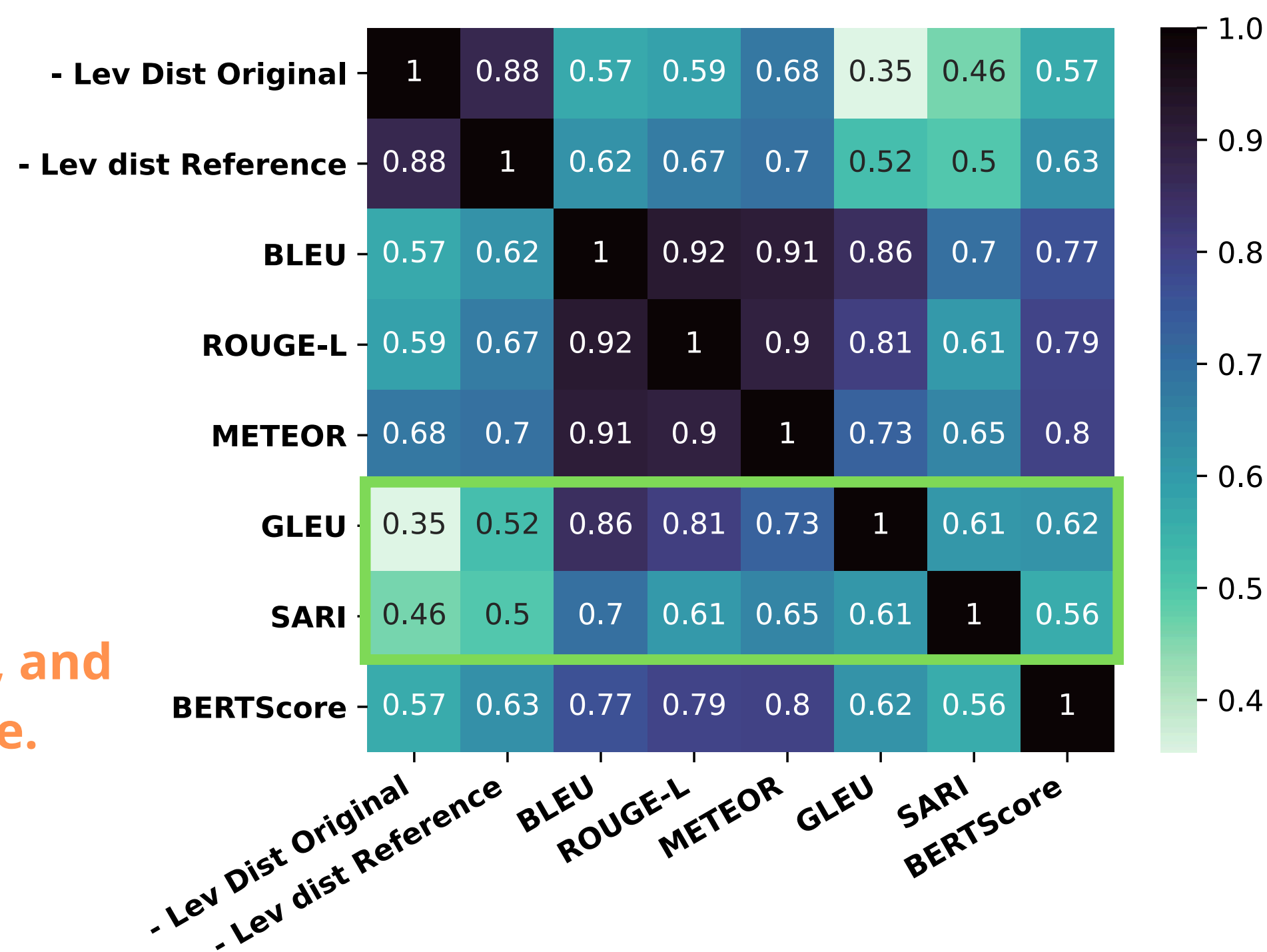
Performances of Revision Models

Revision Model ↓ / Metric →	BLEU	ROUGE-L	METEOR	GLEU	SARI	Bertscore
no edits control approach	66.00	78.30	83.80	25.78	60.63	95.95
CoEdIT-XL	50.24	67.46	66.66	23.84	39.60	93.90
Mistral-7B-Instruct-v0.2	27.77	50.79	54.02	15.38	31.63	92.14
Meta-Llama-3-8B-Instruct	41.66	62.07	62.00	25.78	39.33	93.53
Meta-Llama-3-70B-Instruct	46.78	65.61	67.20	30.31	42.74	93.90
GPT4o-mini	<u>51.68</u>	<u>69.54</u>	<u>72.70</u>	32.67	<u>45.06</u>	<u>94.80</u>
GPT4o	49.34	68.20	69.88	<u>31.35</u>	43.54	94.45

Limitations of Similarity-based Metrics

1. Most metrics are highly correlated, providing redundant information
2. Metrics only capture surface similarity
3. Substantial revisions are penalised

SARI and GLEU stand out as exceptions, and are the only ones to consider the source.



Exploring Alternative Evaluation Approaches

 **Goal:** Define a set of complementary and reliable metrics to evaluate paragraph revision

 **1. Manual pairwise evaluation** of generated revisions by different models

3 levels:

1. **Relatedness** (to the instruction)
2. **Correctness**
3. **Preference**

 **2. Identify candidates metrics**

Related NLP Domains

Input: Original + Generated

BETS: Text simplification

BLANC: Summarization

ParaPLUIE: Paraphrase detection

LLM-as-a-judge

Input: Original + Generated (+ Gold)

LLM-choice

(Yes/No + Pairwise comparison)

LLM-likert

(Generating Scores)

3. Align human judgment to automatic evaluation

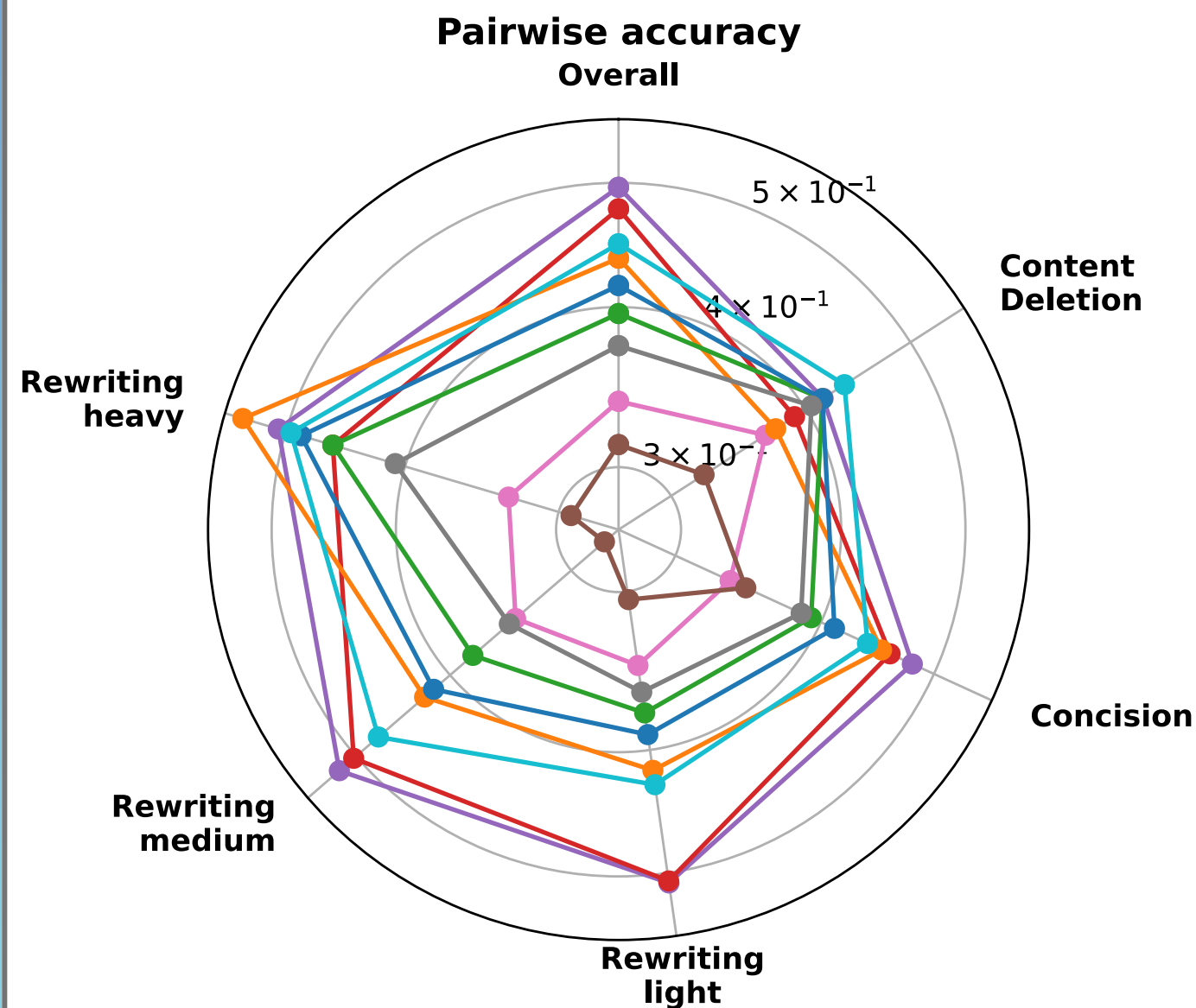
Results

Judge	Pair acc.	V	K
Avg. LLM choice	0.496	0.239	0.247
Avg. LLM likert	0.338	0.240	0.181
ParaPLUIE	0.477	0.225	0.197
BETS	0.437	0.152	0.127
BLANC	0.312	0.117	-0.080
Bertscore	0.395	0.161	0.034
SARI	0.416	0.184	0.071
GLEU	0.448	0.193	0.138
ROUGE-L	0.373	0.179	-0.013
<i>Random</i>	<i>0.264</i>	<i>0.027</i>	<i>-0.006</i>

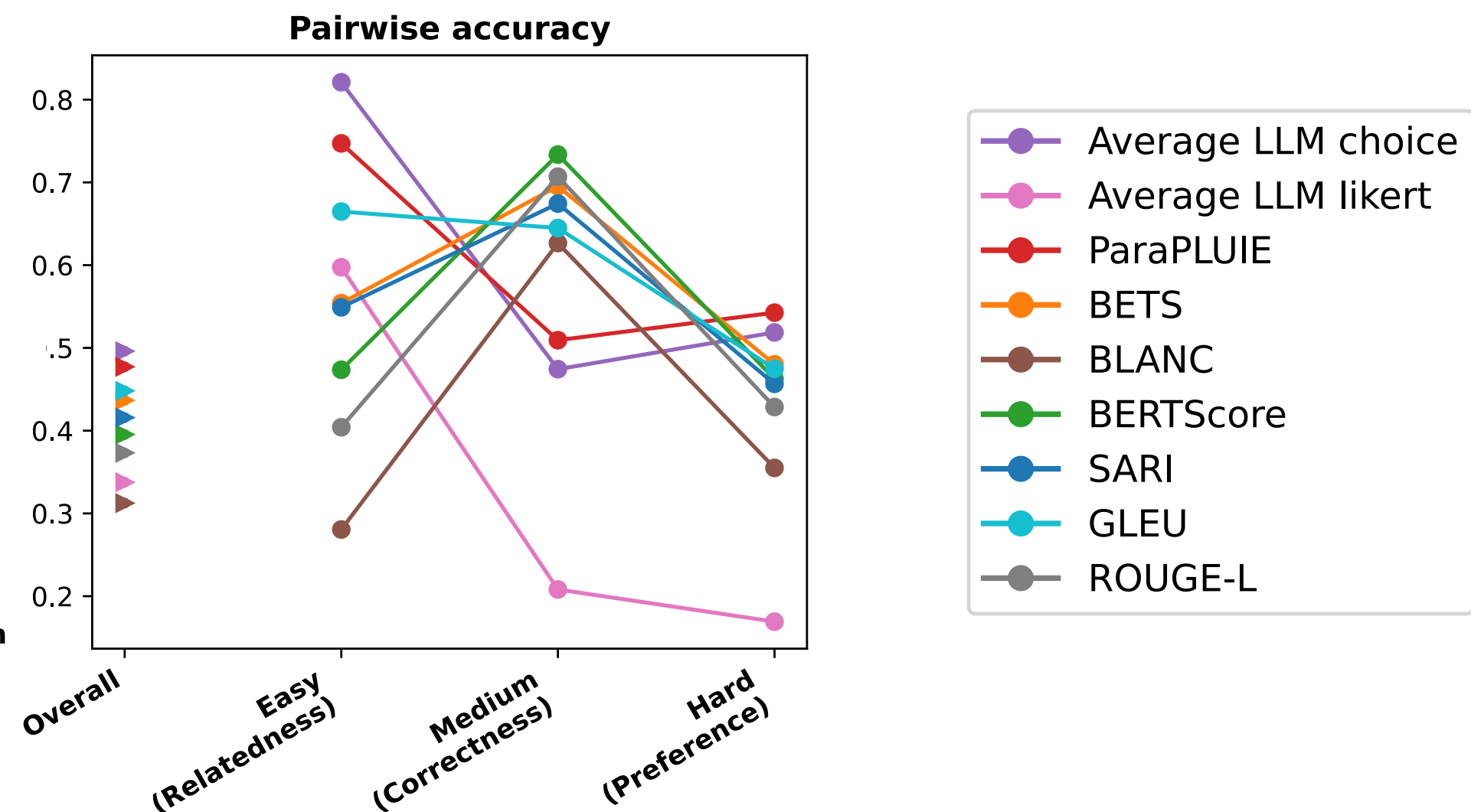
Alignment of automatic metrics with human judgements

Performance by Aspect

By Difficulty



By Revision Category



Proposed Alternative Evaluation

1. **Instruction following**: LLM-as-a-judge
2. **Similarity to gold**: SARI or GLEU
3. **Meaning-preservation**: ParaPLUIE

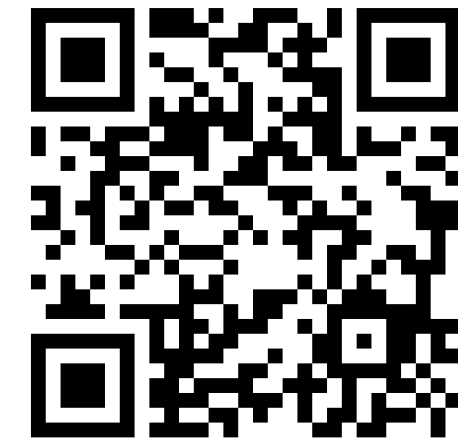
Conclusion

Contributions recap

- Traditional similarity metrics alone **fail to accurately evaluate** text revision
- An **alternative evaluation method** composed of 3 complementary metrics
- **ParaReval, a dataset** of human pairwise evaluations of generated revisions

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Paper:



Appendix: Example

